

材料系 材料力学B 2022春 Quiz

Quiz 其实就是某一年的期中考试题

Quiz of Mechanics of Materials B

Name:

Student Number:

1. The corners B and D of the square plate are given the displacements indicated. Determine the shear strains at A and average normal strain along DB (20 points).

Applying trigonometry to Fig. a

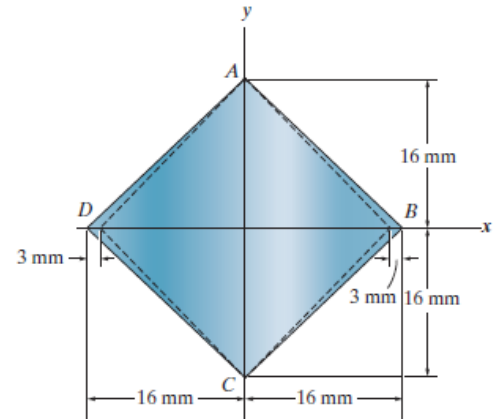
$$\phi = \tan^{-1} \left(\frac{13}{16} \right) = 39.09^\circ \left(\frac{\pi \text{ rad}}{180^\circ} \right) = 0.6823 \text{ rad}$$

By the definition of shear strain,

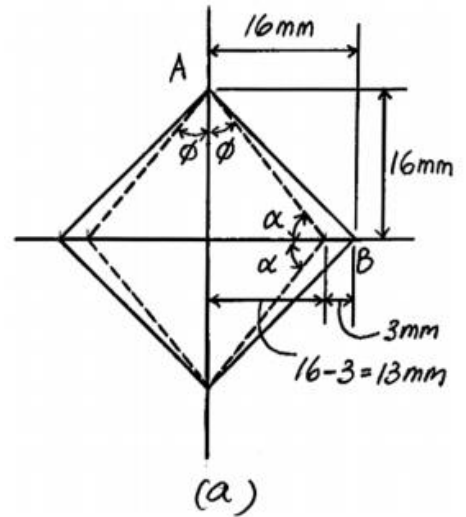
$$(\gamma_{xy})_A = \frac{\pi}{2} - 2\phi = \frac{\pi}{2} - 2(0.6823) = 0.206 \text{ rad}$$

Average normal strain for DB:

$$\epsilon_{avg} = \frac{L_{(BD)'} - L_{BD}}{L_{BD}} = \frac{26 - 32}{32} = -0.1875 \text{ mm/mm} \quad \text{Ans.}$$



Ans.



2. Determine the elongation of the square hollow bar when it is subjected to the axial force of 100kN. If this axial force is increased to 360kN and released, find the permanent elongation of the bar. The bar is made of a metal alloy having a stress–strain diagram which can be approximated as shown (20 points).

Normal Stress and Strain: The cross-sectional area of the hollow bar is $A = 0.05^2 - 0.04^2 = 0.9(10^{-3})\text{m}^2$. When $P = 100\text{kN}$,

$$\sigma_1 = \frac{P}{A} = \frac{100(10^3)}{0.9(10^{-3})} = 111.11 \text{ MPa}$$

From the stress–strain diagram shown in Fig. *a*, the slope of the straight line *OA* which represents the modulus of elasticity of the metal alloy is

$$E = \frac{250(10^6) - 0}{0.00125 - 0} = 200 \text{ GPa}$$

Since $\sigma_1 < 250 \text{ MPa}$, Hooke's Law can be applied. Thus

$$\begin{aligned}\sigma_1 &= E\varepsilon_1; 111.11(10^6) = 200(10^9)\varepsilon_1 \\ \varepsilon_1 &= 0.5556(10^{-3}) \text{ mm/mm}\end{aligned}$$

Thus, the elongation of the bar is

$$\delta_1 = \varepsilon_1 L = 0.5556(10^{-3})(600) = 0.333 \text{ mm}$$

When $P = 360 \text{ kN}$,

$$\sigma_2 = \frac{P}{A} = \frac{360(10^3)}{0.9(10^{-3})} = 400 \text{ MPa}$$

From the geometry of the stress–strain diagram, Fig. *a*,

$$\frac{\varepsilon_2 - 0.00125}{400 - 250} = \frac{0.05 - 0.00125}{500 - 250} \quad \varepsilon_2 = 0.0305 \text{ mm/mm}$$

When $P = 360 \text{ kN}$ is removed, the strain recovers linearly along line *BC*, Fig. *a*, parallel to *OA*. Thus, the elastic recovery of strain is given by

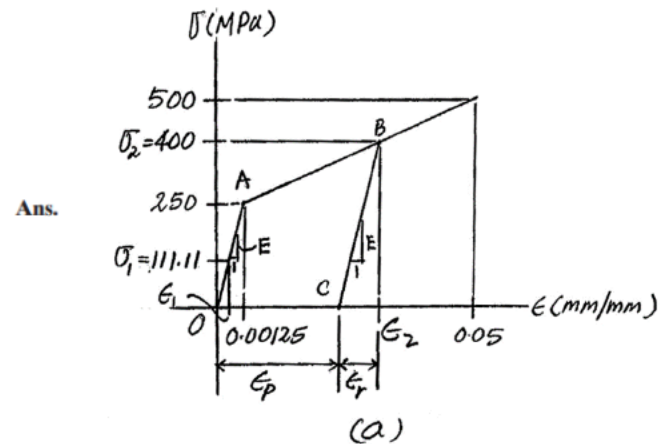
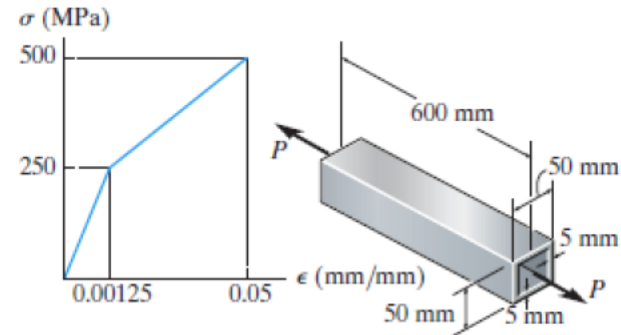
$$\begin{aligned}\sigma_2 &= E\varepsilon_r; 400(10^6) = 200(10^9)\varepsilon_r \\ \varepsilon_r &= 0.002 \text{ mm/mm}\end{aligned}$$

The permanent set is

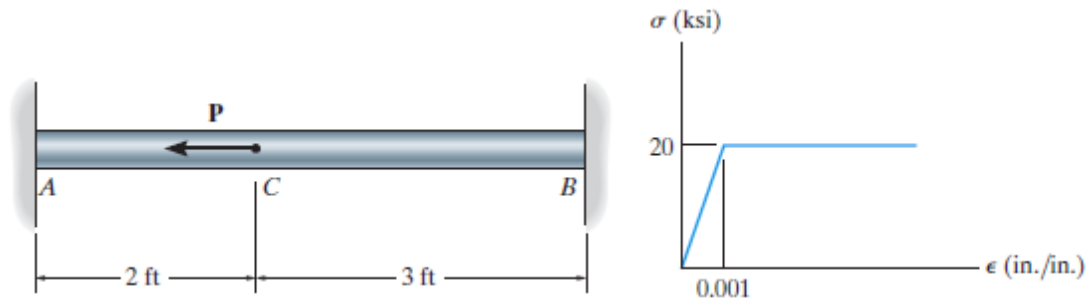
$$\varepsilon_p = \varepsilon_2 - \varepsilon_r = 0.0305 - 0.002 = 0.0285 \text{ mm/mm}$$

Thus, the permanent elongation of the bar is

$$\delta_p = \varepsilon_p L = 0.0285(600) = 17.1 \text{ mm}$$



3. The bar having a diameter of 2 in. is fixed connected at its ends and supports the axial load **P**. If the material is elastic perfectly plastic as shown by the stress–strain diagram, determine the smallest load **P** needed to cause segment **CB** to yield. If this load is released, determine the permanent displacement of point **C** (20points).



When **P** is increased, region **AC** will become plastic first, then **CB** will become plastic. Thus,

$$F_A = F_B = \sigma A = 20(\pi)(1)^2 = 62.832 \text{ kip}$$

$$\rightarrow \Sigma F_x = 0; \quad F_A + F_B - P = 0 \quad (1)$$

$$P = 2(62.832) = 125.66 \text{ kip}$$

$$P = 126 \text{ kip} \quad \text{Ans.}$$

The deflection of point **C** is,

$$\delta_C = \epsilon L = (0.001)(3)(12) = 0.036 \text{ in.} \leftarrow$$

Consider the reverse of **P** on the bar.

$$\frac{F_A'(2)}{AE} = \frac{F_B'(3)}{AE}$$

$$F_A' = 1.5 F_B'$$

So that from Eq. (1)

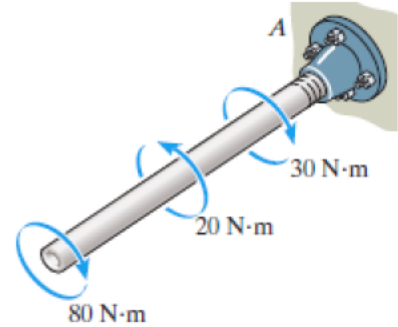
$$F_B' = 0.4P$$

$$F_A' = 0.6P$$

$$\delta_C' = \frac{F_B' L}{AE} = \frac{0.4(P)(3)(12)}{AE} = \frac{0.4(125.66)(3)(12)}{\pi(1)^2(20/0.001)} = 0.02880 \text{ in.} \rightarrow$$

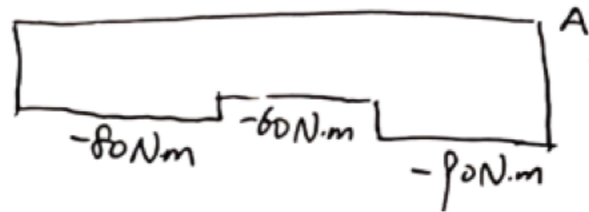
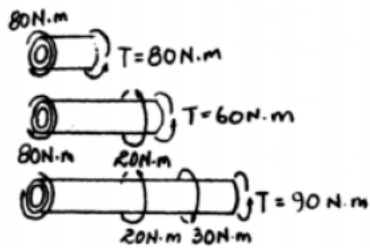
$$\Delta\delta = 0.036 - 0.0288 = 0.00720 \text{ in.} \leftarrow \quad \text{Ans.}$$

4. The copper pipe has an outer diameter of 40 mm and an inner diameter of 37 mm. If it is tightly secured to the wall at A and three torques are applied to it as shown, determine the absolute maximum shear stress developed in the pipe (20 points).

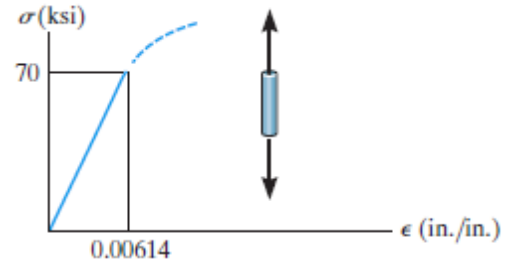


$$\tau_{\max} = \frac{T_{\max} c}{J} = \frac{90(0.02)}{\frac{\pi}{2} (0.02^4 - 0.0185^4)}$$

$$= 26.7 \text{ MPa}$$



5. The elastic portion of the tension stress–strain diagram for an aluminum alloy is shown in the figure. The specimen used for the test has a gauge length of 2 in. and a diameter of 0.5 in. If the applied load is 10 kip, determine the new diameter of the specimen. The shear modulus is ~~38000~~ 3800 ksi (20 points).



$$\sigma = \frac{P}{A} = \frac{10}{\frac{\pi}{4}(0.5)^2} = 50.9296 \text{ ksi}$$

From the stress–strain diagram

$$E = \frac{70}{0.00614} = 11400.65 \text{ ksi}$$

$$\varepsilon_{\text{long}} = \frac{\sigma}{E} = \frac{50.9296}{11400.65} = 0.0044673 \text{ in./in.}$$

$$G = \frac{E}{2(1 + \nu)}; \quad 3.8(10^3) = \frac{11400.65}{2(1 + \nu)}; \quad \nu = 0.500$$

$$\varepsilon_{\text{lat}} = -\nu \varepsilon_{\text{long}} = -0.500(0.0044673) = -0.002234 \text{ in./in.}$$

$$\Delta d = \varepsilon_{\text{lat}} d = -0.002234(0.5) = -0.001117 \text{ in.}$$

$$d' = d + \Delta d = 0.5 - 0.001117 = 0.4989 \text{ in.}$$

Ans.